GPU Infrastructure

Architecture & Cost Overview — Scale-to-Zero GPU Kubernetes on Oracle Cloud

PREPARED FOR CLIENT REVIEW

1 The Problem

Running GPU servers 24/7 is extremely expensive. A single NVIDIA A10 GPU instance costs **~$1,440/month** if left running continuously. For AI workloads that only need GPUs when users are actively making requests, this is a massive waste.

Our solution: A Kubernetes cluster that keeps GPU servers turned off when nobody is using them, and automatically boots them on-demand when a request arrives.

2 Infrastructure Overview

The cluster runs on **Oracle Cloud Infrastructure (OKE)** and consists of two separate node pools:

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CPU Node Pool

Always Running — **~$27/month**

- Shape: VM.Standard.E4.Flex
- 1 OCPU, 8 GB RAM
- 1 node (fixed)

Runs:

- KEDA HTTP Interceptor
- Cluster Autoscaler
- NVIDIA GPU Operator

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GPU Node Pool

Scale-to-Zero — **$2.00/hr only when active**

- Shape: VM.GPU.A10.1
- 1× NVIDIA A10 (24 GB VRAM)
- 0–2 nodes (dynamic)

Runs:

- AI Model (Llama 3 8B)
- GPU Dashboard

CPU Node acts as the "receptionist" — it receives all web requests, decides if a GPU is needed, and manages the scaling process.

GPU Node contains the NVIDIA A10 GPU. It only exists when there are active requests. After 5 minutes idle, it is automatically shut down.

3 How a Request Flows Through the System

1. **User sends request** to gpu.yourdomain.com
2. **Load Balancer** routes traffic to KEDA Interceptor on CPU node
3. **KEDA holds the request** and tells Kubernetes: "Scale GPU pod from 0 → 1"
4. **Cluster Autoscaler** sees a pending GPU pod → tells OCI to create a VM.GPU.A10.1 server
5. **OCI provisions GPU server** (~3–5 minutes)
6. **GPU Operator** auto-installs NVIDIA drivers on the new server
7. **AI pod starts**, processes the request, returns the response
8. **After 5 min idle** → GPU pod scales to 0, GPU server removed, cost drops to **$0**

Scenario	Response Time
Cold start (GPU is off)	~5–10 minutes (first request only)
Warm (GPU already running)	Instant (milliseconds)
Scale down (no activity)	5 min idle → GPU shuts off

> **Note:** The cold start wait only happens for the very first request after idle. Once the GPU is running, all subsequent requests are instant.

4 Key Technologies

Technology	What It Does	Why We Use It
OKE	Managed Kubernetes cluster	Oracle manages the control plane for free
KEDA	Scales pods based on HTTP traffic	Holds requests while pods boot — prevents 502 errors
Cluster Autoscaler	Scales GPU servers from 0 to N	Provisions/removes real GPU hardware on demand
GPU Operator	Installs GPU drivers automatically	No manual intervention when new servers boot
vLLM	LLM inference engine	Serves Llama via OpenAI-compatible API

5 Monthly Cost Estimate

Fixed Costs (Always Running)

Component	Status	Cost
CPU Node (1 OCPU, 8 GB)	24/7	~$27/mo
OKE Control Plane (Basic)	24/7	Free
OCI Load Balancer	24/7	Free

Variable Costs (GPU — On Demand)

Component	Spec	Cost
GPU Node	VM.GPU.A10.1 (1× A10, 24 GB)	**$2.00/hr** (only when active)

Usage Scenarios

Scenario	GPU Hours/Month	GPU Cost	Total/Month
Light (2 hrs/day)	~60 hrs	~$120	**~$147**
Moderate (8 hrs/day, weekdays)	~176 hrs	~$352	**~$379**
Heavy (12 hrs/day)	~360 hrs	~$720	**~$747**
Always On (24/7)	720 hrs	~$1,440	**~$1,467**

Source: Oracle Cloud Compute Pricing

6 💰 Savings With Our Architecture

You only pay for GPU time when the AI is actually processing requests. During nights, weekends, and idle periods, the GPU cost is **$0**.

Your Usage Pattern	You Pay	You Save Per Year
GPU runs 2 hrs/day	**$147/month**	$15,840/year saved (90%)
GPU runs 8 hrs/day (weekdays)	**$379/month**	$13,056/year saved (74%)
GPU runs 8 hrs/day (Mon–Fri)	**$347/month**	$13,440/year saved (76%)
GPU runs 12 hrs/day	**$747/month**	$8,640/year saved (49%)

Compared to a traditional always-on GPU server at $1,467/month ($17,604/year).

Your Likely Scenario — Business Hours (9 AM – 5 PM, Sun–Thu)

🎯 **$13,056 saved/year**

You pay $4,548/year instead of $17,604/year

How It Works

✅ GPU is running	You pay **$2.00/hour**
⏸️ Nobody is using it	GPU shuts off after 5 min → you pay **$0**
🔄 Someone sends a request	GPU boots back up automatically — no manual intervention

7 GPU Hardware: NVIDIA A10

Spec	Value
GPU	NVIDIA A10 Tensor Core
VRAM	24 GB GDDR6
Architecture	Ampere
Best for	AI inference, Llama 3 8B, image generation
OCI Shape	VM.GPU.A10.1

AI Model Compatibility

Model	VRAM Required	Fits on A10?
Llama 3.1 8B (16-bit)	~16 GB	✅ Yes
Llama 3.1 8B (4-bit quantized)	~6 GB	✅ Yes — fastest
Llama 2 13B (8-bit quantized)	~14 GB	✅ Yes
Llama 3.1 70B	~140 GB	❌ No — needs 8× A100

8 Security

Aspect	Implementation
Worker Nodes	Private subnet (no public IP)
API Endpoint	Public (for management via kubectl)
Cluster Type	OKE Basic (Oracle-managed control plane)
Secrets	Oracle-managed encryption
GPU Drivers	Auto-installed via GPU Operator (no SSH needed)
Dashboard	Disabled (kubectl only — reduces attack surface)

9 Summary

- ✅ GPU scales to zero when idle
- ✅ Automatic GPU provisioning on request
- ✅ No 502 errors during cold start
- ✅ Automatic GPU driver installation
- ✅ Ready for Llama 3 8B inference
- ✅ Monthly cost ~$147 (light usage)
- ✅ Up to 90% savings vs always-on
- ✅ Private, secure architecture